

Full Moment Tensor Analysis of Nuclear Explosions in North Korea

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ABSTRACT

A seismic moment tensor is a 3×3 symmetric matrix that characterizes the far-field radiation from a seismic source such as an earthquake or an explosion. We estimate full (unconstrained) moment tensors and their uncertainties for seven events at the North Korea nuclear test site. Six of the events were declared nuclear tests, and the remaining one, which we interpret as a cavity collapse, was an event that occurred eight minutes after the 2017 declared test. We also analyze two earthquake events that occurred to the south of the site. For each of the nine events, we perform a grid search over the 6D space of moment tensors, generating synthetic waveforms at each moment tensor grid point and then minimizing a misfit function between the observed and synthetic waveforms. We characterize the uncertainty of each inferred moment tensor in three ways: We use the variation in waveform misfit on the eigenvalue lune, we use a probability density function for moment tensor source types, and we use a confidence curve. We find that the source types (eigenvalue triples) for the nine events separate into three distinct groups, with one group for the nuclear tests, one group for the two earthquakes, and one group just for the collapse. We find that each of the six moment tensors for the declared tests is the sum of a double couple and a crack tensor, with the crack plane being nearly horizontal.

Electronic Supplement: Figures showing the regimes for moment tensors, waveform fits, uncertainty analyses, and time shifts for Rayleigh and Love waves.

INTRODUCTION

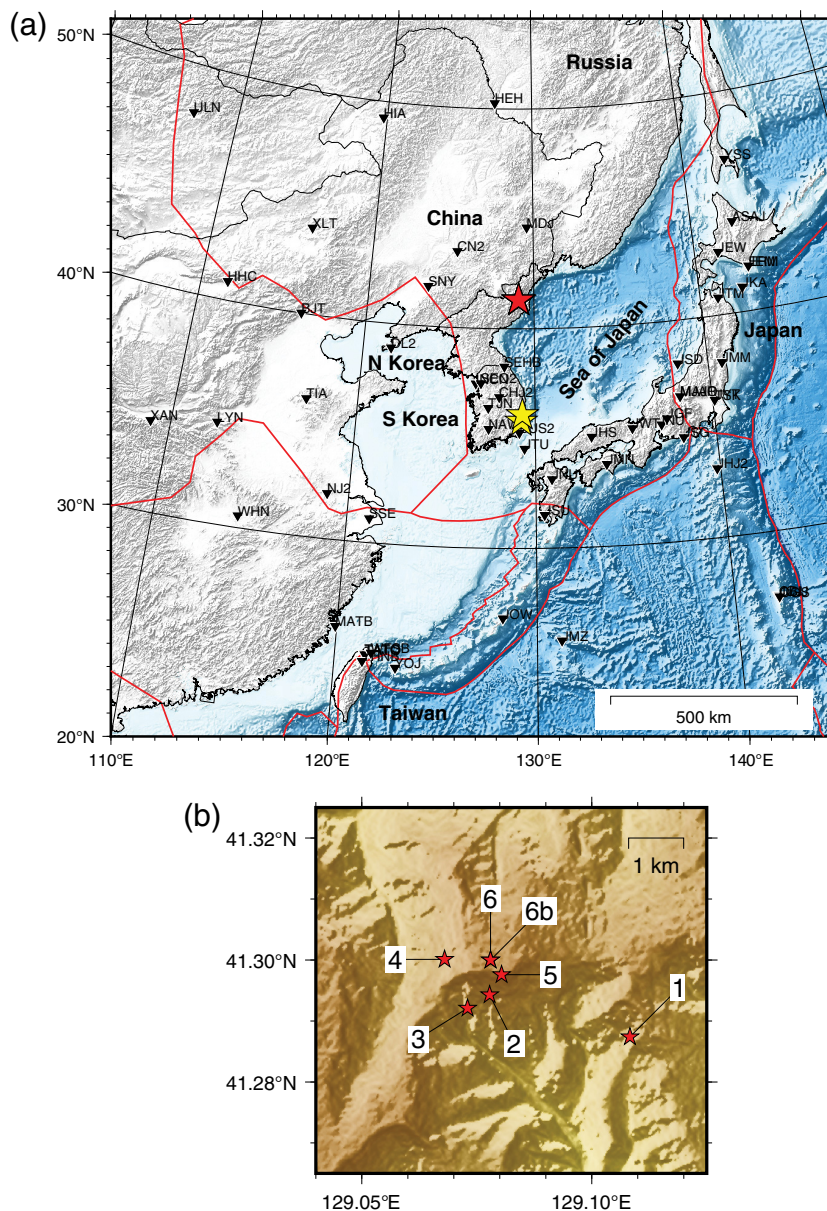
Seismic moment tensors have proven valuable for discriminating explosion events from earthquakes and collapse events (Ford *et al.*, 2009a). Estimating a moment tensor for a given event involves comparing recorded seismograms with synthetic seismograms computed using a 3×3 symmetric matrix to represent the earthquake source and using an assumed model of Earth's seismic structure. Between October 2006 and September 2017, North Korea (Democratic People's Republic of Korea) declared six nuclear tests (Table 1). These events were

recorded by regional seismic stations and provide a rare set of events for examining the estimation of (full) seismic moment tensors (Ford *et al.*, 2009b; Barth, 2014; Vavryčuk and Kim, 2014; Cesca *et al.*, 2017; Ichinose *et al.*, 2017; Liu *et al.*, 2018).

We estimate moment tensors, with uncertainties, for all six explosion events as well as for three events: one cavity collapse event and two earthquakes. With inclusion of first-motion polarities, careful treatment of time shifts between data and synthetics, and careful selection of the bandpass for processing waveforms, we obtain moment tensor source types that clearly separate among explosions, earthquakes, and implosions. We interpret the moment tensors in terms of a one-process model of an oblique opening across a fault plane and as a two-process model of a tensional crack plus a shear fault aligned with the crack plane. For the six explosion events, we find that all six events can be described in terms of the two-process crack-plus-double-couple (CDC) model (Minson *et al.*, 2007; Tape and Tape, 2013), with the crack tensor having its plane oriented horizontally. The emphasis in our article is in conveying and visualizing the uncertainties in full moment tensors (Ford *et al.*, 2010; Stähler and Sigloch, 2014), following the approach of Alvizuri *et al.* (2018), and in documenting how choices in the estimation problem affect uncertainties.

METHODS

Our moment tensor inversion method involves performing a complete search over the parameter space of moment tensors (including magnitude) and depth. We use an adapted version of the cut-and-paste code (Zhao and Helmberger, 1994; Zhu and Helmberger, 1996; Zhu and Ben-Zion, 2013) that has been applied recently for uncertainty quantification (Alvizuri and Tape, 2016; Silwal and Tape, 2016; Alvizuri *et al.*, 2018). For each moment tensor in the parameter space, synthetic seismograms are computed using a frequency–wavenumber approach (Zhu and Rivera, 2002) with a 1D (layered) Earth model, and then these seismograms are compared with observed seismograms via a misfit function. For details, see Alvizuri *et al.* (2018); next, we discuss details specific to this study.



▲ **Figure 1.** Locations of events and stations in this study. (a) Map of North Korea and surrounding region, including seismic stations used in this study. The North Korea test site is marked with a red star. The two yellow stars denote earthquake epicenters in this study. Plate boundaries (red) are from [Bird \(2003\)](#). (b) Zoom-in view of the North Korea test site. Locations are from Table 1. The color version of this figure is available only in the electronic edition.

Within the Global Centroid Moment Tensor catalog there are 13 crustal (depth ≤ 30 km) earthquakes within 700 km of the North Korea nuclear test site that occurred between 1 January 2009 and 1 January 2018. We choose a pair of events, on 12 September 2016 and 15 November 2017, which occurred to the south on the Korean peninsula (Fig. 1a). These events provide an opportunity to compare nonexplosion events in the same region—and recorded by the same stations—as the explosion events. Table 1 lists hypocenters, origin times, and magnitudes obtained from other studies.

For computing synthetic seismograms, we use the same four-layered model as listed in table 1 of [Ford et al. \(2009b\)](#). This model, labeled MDJ2, was based on the MDJ model of [Nguyen \(1994\)](#) and modified by [Ford et al. \(2009b\)](#) to account for large time shifts required for stations BJT and HIA, at distances > 1000 km to the west and northwest, respectively (Fig. 1a). MDJ is a seismic station in Mudanjiang, China, north of the target events.

RESULTS

Our moment tensor inversion results for all nine events are presented in Tables 2 and 3 and Figures 2–5. [Alvizuri \(2018\)](#) contains an expanded version of waveform fits and uncertainty figures for all events. The choice of band-pass and stations can significantly impact the estimation of moment tensors, including magnitudes. Therefore, we perform two sets of inversions of all nine events: (1) using the same bandpass (20–50 s) and same four stations (IC.MDJ, IU.MAJO, G.INU, and IC.BJT) and (2) using a variable bandpass and all possible stations. Moment tensor source types are represented on the eigenvalue lune ([Riedesel and Jordan, 1989](#); [Tape and Tape, 2012](#); © Fig. S1, available in the electronic supplement to this article).

Six Explosion Events in North Korea

For the six explosion events, the results for the four-station inversions are shown in Figures 2 and 3 and listed in Table 2. The results for the all-station inversion of event NK6 are shown in Figure 4 and © Figure S2; results for other events are shown in [Alvizuri \(2018\)](#). For the four-station and all-station cases and for all six events, the best-fitting moment tensors are mechanisms whose vector fields are radially outward. These mechanisms plot above the λ_3 arc on the lune (© Fig. S1).

First-motion polarities have proven useful for reducing the uncertainty of waveform-based estimations of full moment tensors ([Chiang et al., 2014, 2016](#); [Alvizuri and Tape, 2016](#); [Wang et al., 2018](#)). For event NK6, we identified first-motion polarities at 11 stations (© Fig. S2a); other stations exhibited emergent or unclear first arrivals. A future effort could incorporate regional and teleseismic first-motion polarities for all events.

Three Nonexplosion Events

Results for the three nonexplosion events, using all stations, are shown in Figure 5; results using four stations are shown in © Figure S3. We obtain an implosion mechanism for NK6b,

Table 1
Event Origin Times, Hypocenters, and Magnitudes

Origin Time						
Event ID	(yyyy/mm/dd hh:mm:ss.ssss)	Longitude (°)	Latitude (°)	Depth (km)	Magnitude	Reference
NK1	2006/10/09 01:35:28.0000	129.1083	41.2874	— (1)	$M_s 2.92 \pm 0.20$	Zhao <i>et al.</i> (2017)
NK2	2009/05/25 00:54:43.1239	129.0778	41.2943	— (1)	$M_s 3.65 \pm 0.07$	Zhao <i>et al.</i> (2017)
NK3	2013/02/12 02:57:51.2725	129.0730	41.2921	— (1)	$M_s 3.94 \pm 0.16$	Zhao <i>et al.</i> (2017)
NK4	2016/01/06 01:30:00.9635	129.0680	41.3001	— (1)	$M_s 4.05 \pm 0.08$	Zhao <i>et al.</i> (2017)
NK5	2016/09/09 00:30:01.3857	129.0804	41.2976	— (1)	$M_s 4.23 \pm 0.09$	Zhao <i>et al.</i> (2017)
NK6*	2017/09/03 03:38:31.8100	129.078	41.300	— (1)	$M_w 5.24$	Wang <i>et al.</i> (2018)
NK6b*	2017/09/03 03:30:01.7600	129.078	41.300	— (1)	$M_w 4.5$	Wang <i>et al.</i> (2018)
EQ1†	2016/09/12 11:32:55.7700	129.2162	35.7808	13 (13)	$M_w 5.4$	USGS
EQ2†	2017/11/15 05:29:32.8200	129.2800	36.0735	10 (3)	$M_s 5.5$	USGS

The depth in parentheses is the one used in our moment tensor inversions.

*Origin times from U.S. Geological Survey (USGS), hypocenters from Wang *et al.* (2018).

†Origin times and hypocenter from USGS.

interpreted as a cavity collapse occurring 8 min after the NK6 explosion (Liu *et al.*, 2018; Wang *et al.*, 2018). For the two events to the south (Fig. 1a), we obtain moment tensors that are close to the double couple (Fig. 5b,c). A grid search over depth (⊕ Fig. S4) reveals stability of the moment tensor source type for the first event (EQ1: 12 September 2016) and instability for the second event (EQ2: 15 November 2017). Our best-fitting depth of 4 km for EQ2 is close to the value of 4.0–4.5 km reported in Grigoli *et al.* (2018). For EQ2, our preferred depth is 3 km, which provides a close-to-double-couple moment tensor, but it is possible that the event has a significant negative isotropic component, as indicated by the source type with maximum probability density (Fig. 5c). Our analysis suggests the possibility of a closing (implosion) component that may be related to the assumed induced process for this event (Grigoli *et al.*, 2018; Kim *et al.*, 2018).

DISCUSSION OF EXPLOSION EVENTS

The best-fitting moment tensors are plotted on the lune in Figure 6, which shows clear separation of the explosions, earthquakes, and collapse. Our discussion will focus on the six explosion events.

Our discussion requires some basic understanding of seismic moment tensors and the conventions used to plot them. A moment tensor is a 3×3 symmetric matrix. It can be depicted as a vector field showing the action of a moment tensor on each point on the unit sphere. Each vector can be decomposed into two components that are tangential and radial to the sphere (Shearer 2009, fig. 9.8). The classical focal mechanism plot used to represent moment tensors is colored solid where the radial component of the vector field is outward, and it is left white where the radial component is inward. Based on the three eigenvalues of the moment tensor, there are four possible coloring patterns, depicted in ⊕ Figure S1: all-colored, all-white, a white equatorial band separating two colored caps,

and a colored equatorial band separating two white caps. The boundary curves separating the colored and white regions are where the radial component of the vector field is zero. Therefore, the coloring of the focal mechanism plots provides a visual guide to the moment tensor vector field.

Consistency of Results for Explosion Events

The six explosion events have a remarkably consistent pattern of waveform misfit over the space of moment tensor source types, that is, the lune (Fig. 3). The pattern exhibits a swath of low-misfit moment tensors that range from the upper left of the lune and down to the lower right, with the best-fitting moment tensor in the all-radially-outward ($\lambda_3 > 0$) region of the lune. This swath is visible when using few stations or all stations.

The swath of well-fitting moment tensors occurs because surface waves radiate from the source at shallow angles and therefore can only constrain the perimeter of the source mechanism (Ekström and Richards, 1994; Shuler *et al.*, 2013). The steeper, downward take-off angles, marked by the piercing points in Figure 4g, represent body waves, which may or may not be used in a given inversion. Figure 4d and ⊕ Figure S5d show the best-fitting focal mechanism plots within the swath of low misfit. We see that they all fall within the half of the lune marked by $\lambda_2 > 0$ (⊕ Fig. S1), representing either all-colored focal mechanism plots ($\lambda_3 > 0$) or white caps with colored bands ($\lambda_2 > 0 > \lambda_3$). Furthermore, the normal to the white caps is approximately upward, leaving the bands on the perimeter of focal mechanism plots.

Interpretations of Full Moment Tensors

The best-fitting moment tensors for all six explosions fall within the all-radially-outward regime ($\lambda_3 > 0$) of the lune (⊕ Fig. S1). In this regime, small changes in a moment tensor's source type—that is, its position on the eigenvalue lune—can lead to large

Table 2
Estimated Moment Tensors for All Nine Events in This Study

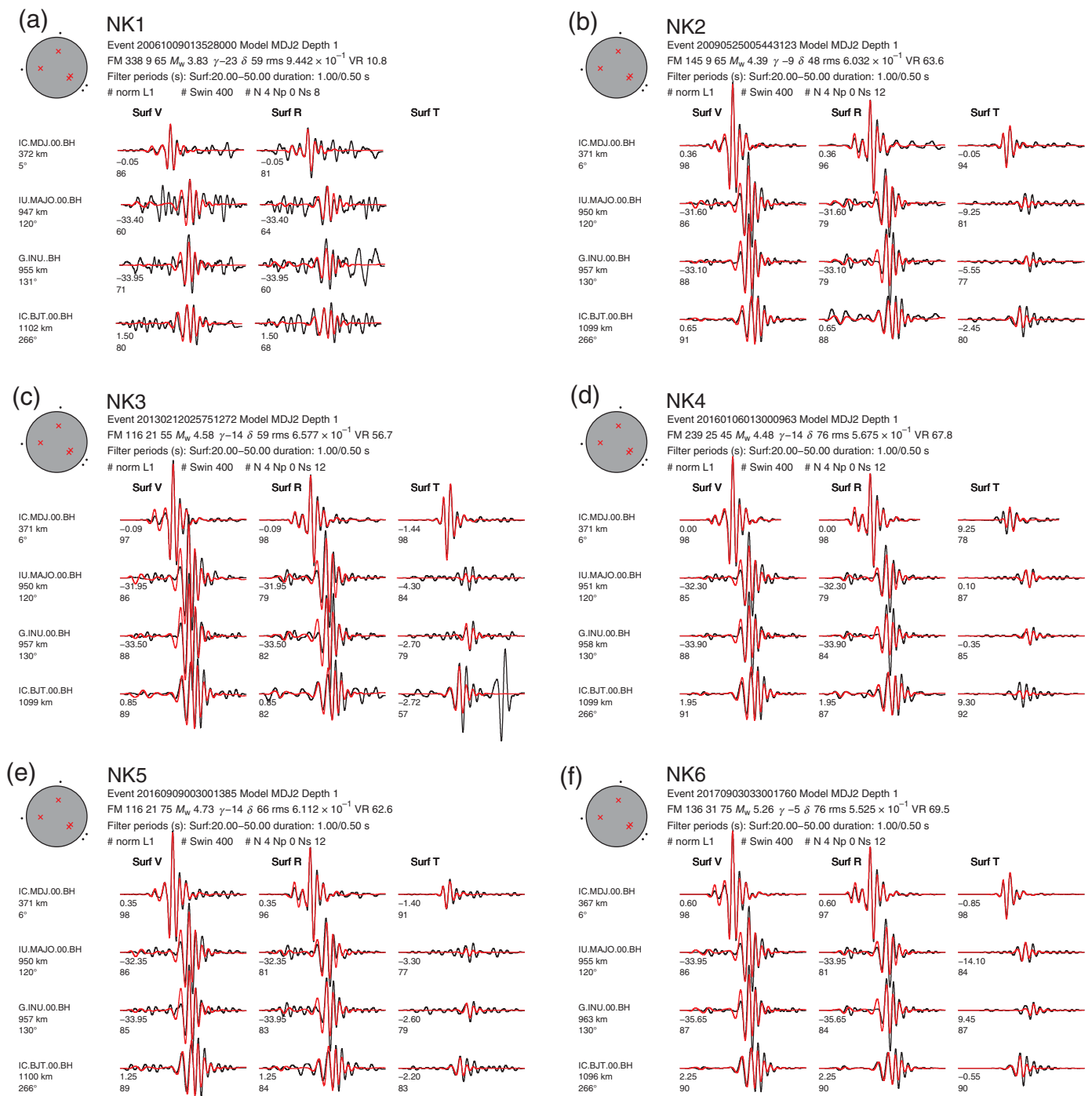
Event	Stations Used	Bandpass Periods (s)	Strike κ	Dip θ	Slip σ	M_w	Lune Longitude γ	Lune Latitude δ	Poisson Ratio ν	Slip Angle α	Lune Azimuth ϕ	Crack Fraction ζ	ν	w	P_{av}
NK1	4	20–50	338	10	65	3.83	–23	59	0.29	43	13	71	–0.31	1.16	0.62
NK2	4	20–50	145	10	65	4.39	–9	48	0.36	74	8	51	–0.15	1.11	0.76
NK3	4	20–50	116	21	55	4.58	–14	59	0.36	65	8	63	–0.22	1.16	0.67
NK4	4	20–50	239	25	45	4.48	–14	76	0.44	65	3	78	–0.22	1.18	0.72
NK5	4	20–50	116	21	75	4.73	–14	66	0.40	65	6	69	–0.22	1.17	0.72
NK6	4	20–50	136	32	75	5.26	–5	76	0.48	82	1	76	–0.08	1.18	0.69
NK6b	4	20–50	137	40	37	4.26	–10	–62	0.60	73	175	64	–0.16	–1.17	0.79
EQ1	4	20–50	52	40	55	5.50	27	13	–4.83	154	–63	68	0.33	0.44	0.97
EQ2	4	20–50	150	64	55	5.50	–23	0	–1.00	43	90	51	–0.31	0.00	0.85

The bandpass and number of stations are fixed for each event; compare with Table 3.

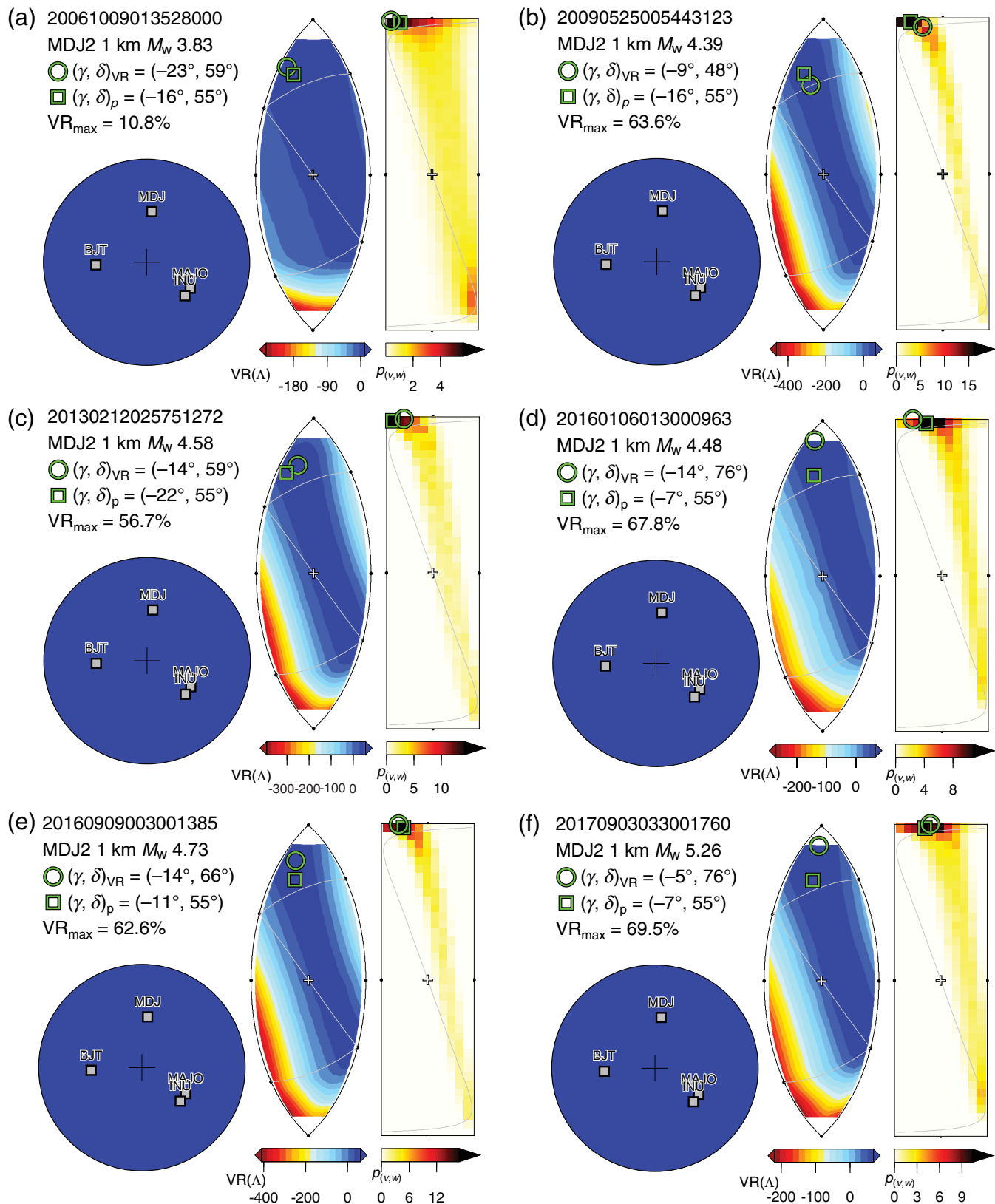
Table 3
Estimated Moment Tensors for All Nine Events in This Study

Event	Stations Used	Bandpass Periods (s)	Strike κ	Dip θ	Slip σ	M_w	Lune Longitude γ	Lune Latitude δ	Poisson Ratio ν	Slip Angle α	Lune Azimuth ϕ	Crack Fraction ζ	ν	w	P_{av}
NK1	18	17–33	22	86	85	3.84	–18	53	0.29	55	13	62	–0.27	1.14	0.58
NK2	21	17–50	126	16	65	4.38	–14	59	0.36	65	8	63	–0.22	1.16	0.63
NK3	16	17–50	284	10	65	4.45	5	53	0.57	98	–3	54	0.08	1.14	0.59
NK4	29	17–50	106	85	–85	4.25	–14	66	0.40	65	6	69	–0.22	1.17	0.71
NK5	34	17–50	150	81	75	4.48	–18	66	0.36	55	8	72	–0.27	1.17	0.62
NK6	34	17–50	328	16	65	5.18	–5	66	0.46	82	2	66	–0.08	1.17	0.80
NK6b	25	25–50	86	37	15	4.20	23	–87	0.48	137	–179	88	0.31	–1.18	0.73
EQ1	32	20–50	116	89	35	5.10	9	–3	–0.50	106	–110	19	0.15	–0.11	0.97
EQ2	29	20–50	333	42	25	5.20	0	–3	0.50	90	180	3	0.00	–0.11	0.86

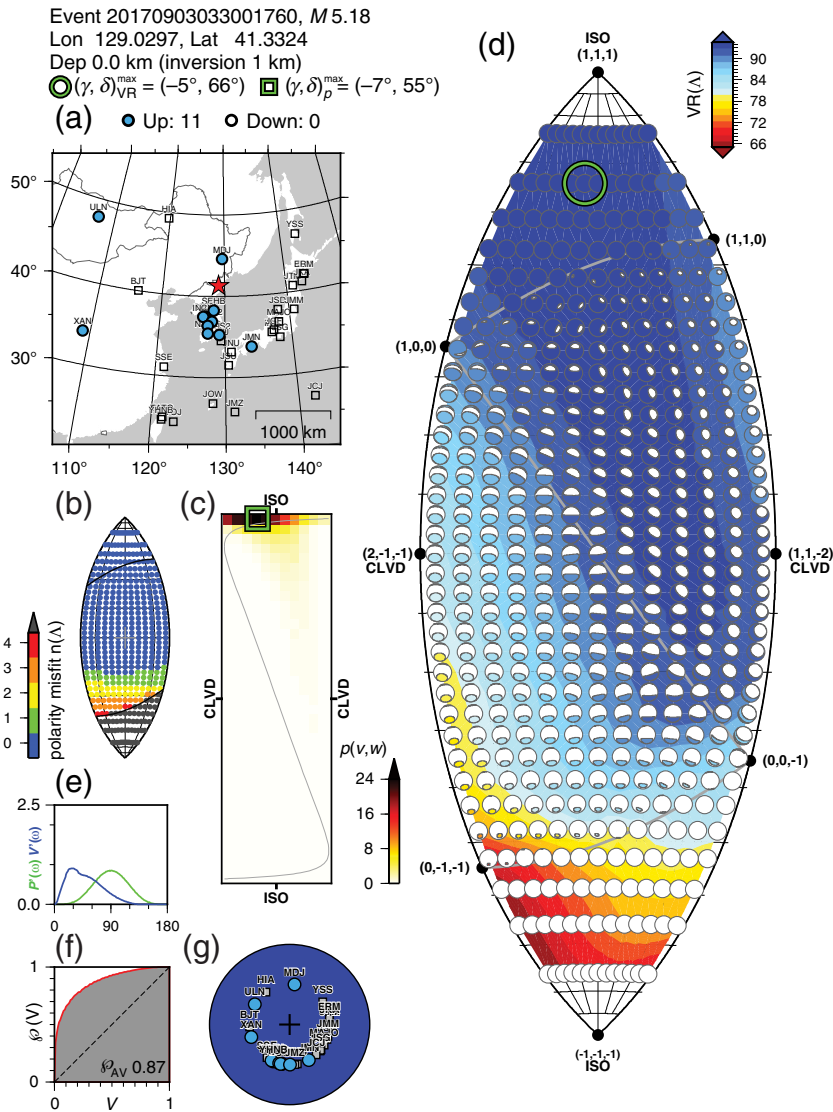
The bandpass and number of stations vary for each event; compare with Table 2.



▲ **Figure 2.** Fits between observed (black) and synthetic (red) waveforms for six North Korea explosion events. For each event, the same bandpass (20–50 s) and the same four stations were used: IC.MDJ, IU.MAJO, G.INU, and IC.BJT. For each station, the three time windows are for Rayleigh waves (Surf V, Surf R) and for Love waves (Surf T). Numbers beneath each waveform pair are the time shift and the cross-correlation maximum. See also Figure 3. (a) NK1: 9 October 2006 M_w 3.84; (b) NK2: 25 May 2009 M_w 4.38; (c) NK3: 12 February 2013 M_w 4.45; (d) NK4: 6 January 2016 M_w 4.25; (e) NK5: 9 September 2016 M_w 4.48; (f) NK6: 3 September 2017 M_w 5.18. The color version of this figure is available only in the electronic edition.



▲ **Figure 3.** Uncertainty analysis for six North Korea explosion events. The focal mechanism plot is the best-fitting moment tensor, indicated by the green circle on the lune plot. The lune shows the variance reduction $VR(\Delta)$; see also ⑤ Figure S1 (available in the electronic supplement to this article). The vw rectangle shows $p(v, w)$, the probability density for source type; the maximum of $p(v, w)$ is denoted by the green square. The waveform fits corresponding to the best-fitting moment tensor (green circle) are shown in Figure 2. The color version of this figure is available only in the electronic edition.

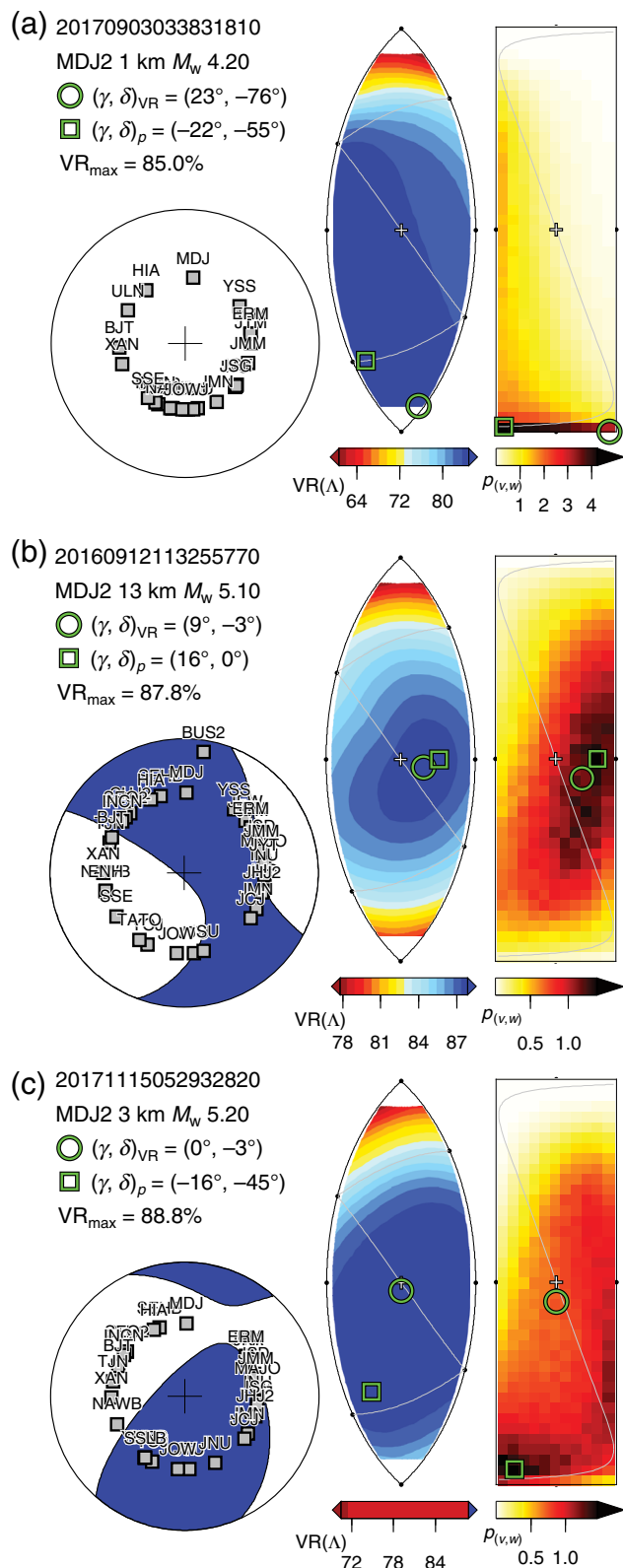


▲ **Figure 4.** Full moment tensor uncertainty summary for the 3 September 2017 M_w 5.18 explosion event. This representation follows [Alvizuri et al. \(2018\)](#) and is shown for other events by [Alvizuri \(2018\)](#). (a) Map of source location (red star) and stations used in the inversion for this event. The station is colored blue if the observed first-motion polarity on the vertical component is up (compression) and white if it is down (dilatation). (b) Contour plot of the polarity misfit on the lune. (c) Source type probability density $p(v, w)$ in the vw rectangle ([Tape and Tape, 2015](#)). A green square indicates the location of the point (v_x, w_x) , in which p is maximum; this point is apt to differ from the source type (v_0, w_0) of M_0 . (d) Contour plot of the $VR(\Delta)$. At each point Δ , the $VR(\Delta)$ is the maximum $VR(M)$ for moment tensors M that have source type Δ . Large values (blue) of VR represent better fit between observed and synthetic waveforms. Of the focal mechanism plots $M(\Delta)$, our solution M_0 (green circle) is the one with largest VR . The gray arcs on the lune are the great circle arcs $\lambda_1 = 0$, $\lambda_2 = 0$, and $\lambda_3 = 0$. Selected eigenvalue triples (black dots) on the boundary of the lune are indicated, with the understanding that the triples need to be normalized. The positive isotropic source $(1, 1, 1)$ is at the top, the negative isotropic source $(-1, -1, -1)$ is at the bottom, and the double couple $(1, 0, -1)$, not shown, would be at the center of the lune. (e) The curves $\hat{V}(\omega)$ and $\hat{P}(\omega)$ that are used to construct the confidence curve $P(V)$ in (f), as explained by [Silwal and Tape \(2016\)](#) in the context of double couples. For full moment tensors, as here, $\hat{V}(\omega) \propto \sin^4 \omega$. (f) The confidence curve $P(V)$ for M_0 . The more the curve resembles the shape of a capital gamma (Γ), the better. The shaded area is the average confidence $\mathcal{P}_{AV}$. (g) The moment tensor M_0 , plotted in a lower-hemisphere projection. The location of the piercing point for each station depends on the station azimuth, the epicentral distance, and the assumed layered reference model. The color version of this figure is available only in the electronic edition.

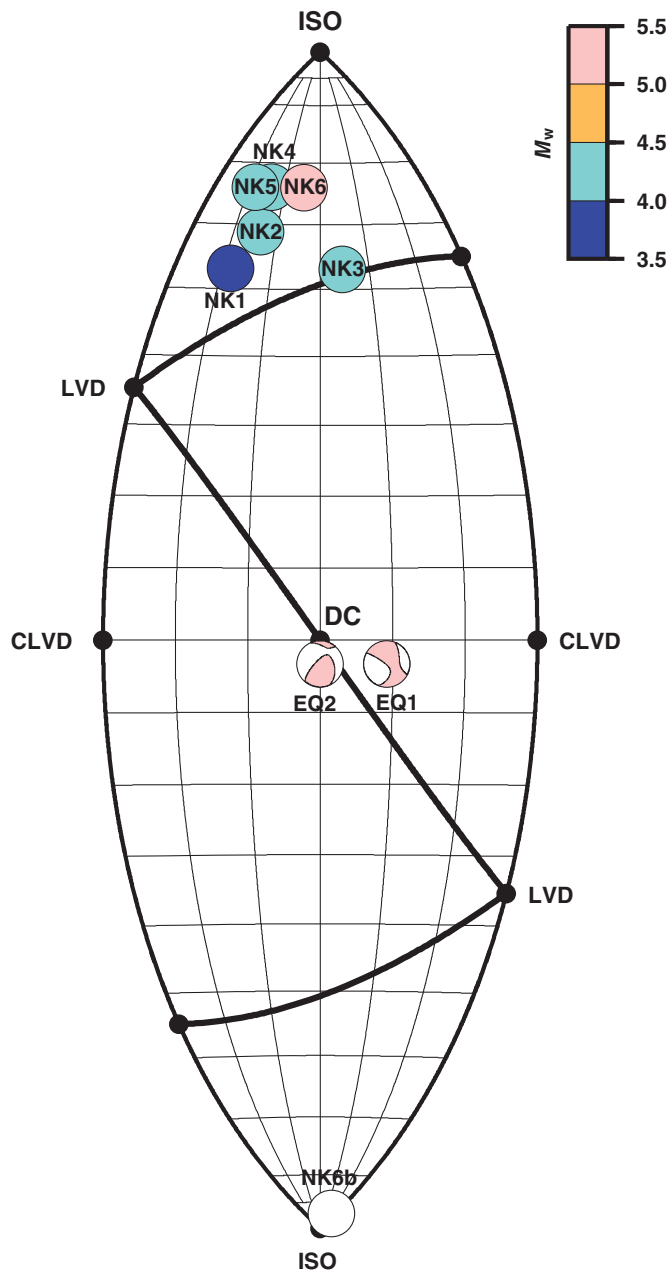
changes in other quantities derived from the moment tensor such as the angle between fault planes.

Figure 7 summarizes three representations of our full moment tensors. Figure 7a shows the six best-fitting moment

tensors for NK1–NK6. The simplistic plotting scheme for focal mechanism plots does not reveal the variations in amplitudes in the moment tensor vector field; these amplitudes can be large, as shown in the representations in Figure 8. Above



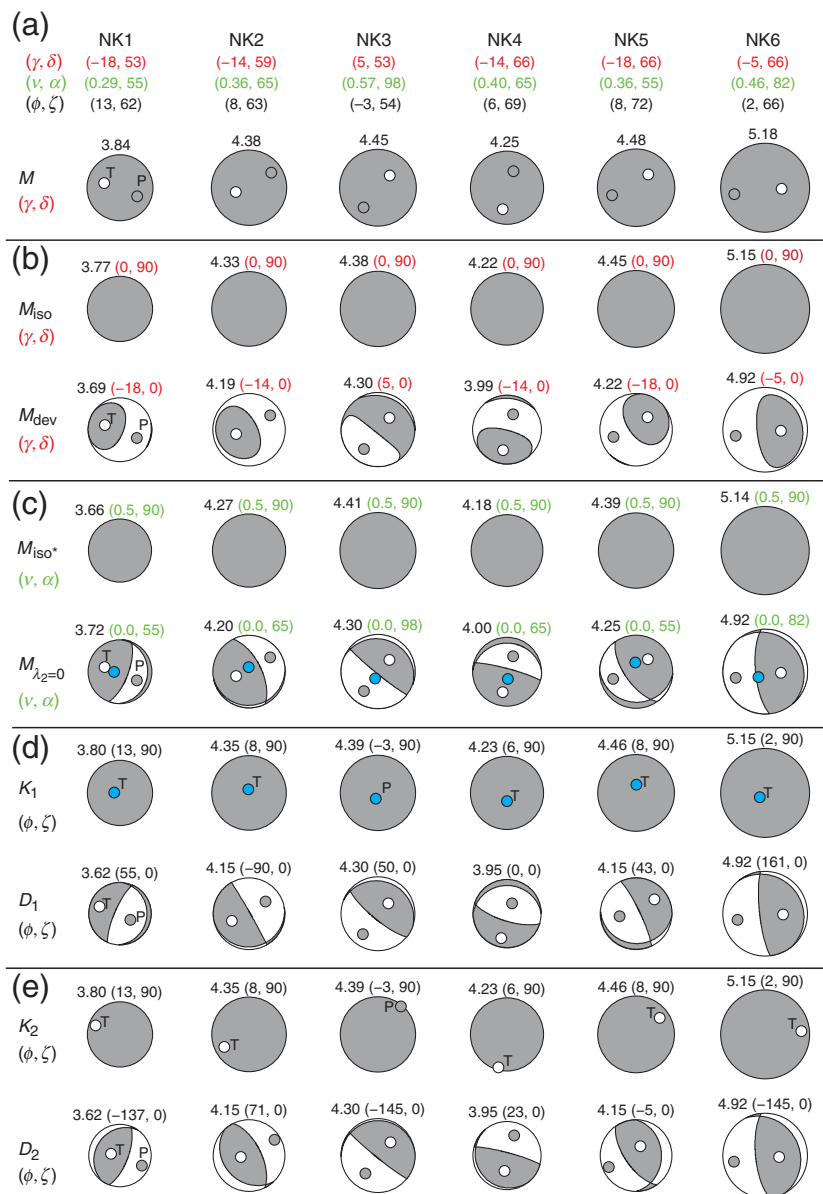
▲ **Figure 5.** Uncertainty analysis for the three nonexplosion events: (a) 3 September 2017 M_w 4.20 collapse (NK6b), (b) 12 September 2016 M_w 5.1 earthquake (EQ1), and (c) 15 November 2017 M_w 5.2 earthquake (EQ2). See waveform fits and expanded versions of the uncertainty analysis in [Alvizuri \(2018\)](#). The color version of this figure is available only in the electronic edition.



▲ **Figure 6.** Best-fitting moment tensors for six explosion events (top), two earthquakes (middle), and the cavity collapse event (NK6b) (bottom). The solid regions of each moment tensor are colored by magnitude; the magnitude of the collapse event is M_w 4.26. Source parameters are listed in Table 3. See [Figure S1](#) for context. The color version of this figure is available only in the electronic edition.

each focal mechanism plot, we list three different pairs of coordinates related to the moment tensor source type (or, specifically, normalized eigenvalues): (1) lune point (γ, δ) , (2) (ν, α) , and (3) (ϕ, ζ) . These values are directly derived from each moment tensor's eigenvalues (Table 3). They also motivate the representations shown in Figure 7b–d.

The first representation Figure 7b shows the isotropic and deviatoric parts of the moment tensor. The P-T axes of the



▲ **Figure 7.** Decompositions of full moment tensors for the six explosion events; see the [Discussion of Explosion Events](#) section for details and also [Figure S6](#). The magnitude is labeled above each focal mechanism plot. (a) The six best-fitting full moment tensors for events NK1, ..., NK6 (Table 3). The colored labels correspond to (γ, δ) (red), (ν, α) (green), (ϕ, ζ) (black) (Table 3). (b) $M = M_{\text{dev}} + M_{\text{iso}}$, in which $M_{\text{iso}} = \frac{1}{3} \text{tr}(M)I$. (c) $M = M_{\lambda_2=0} + M_{\text{iso}*}$, in which $M_{\text{iso}*} = \sqrt{3}\lambda_2 I$. The label (ν, α) above is the Poisson parameter ν and the angle α between fault planes (Aki and Richards, 1980; Tape and Tape, 2013). The blue dot is the normal vector to the near-horizontal fault plane. (d) $M = K_1 + D_1$: one of two decompositions for the crack-plus-double-couple (CDC) model, in which K_1 is the crack tensor and D_1 is the double-couple moment tensor (Minson *et al.*, 2007; Tape and Tape, 2013). The label (ϕ, ζ) above each focal mechanism plot is the lune azimuth to K and ζ is the crack fraction of M . (e) $M = K_2 + D_2$: the second possibility for the CDC model. The color version of this figure is available only in the electronic edition.

deviatoric tensor are the same as those for the starting moment tensor (Fig. 7a). The labels above the moment tensors in Figure 7a,b show the moment magnitude M_w and the lune

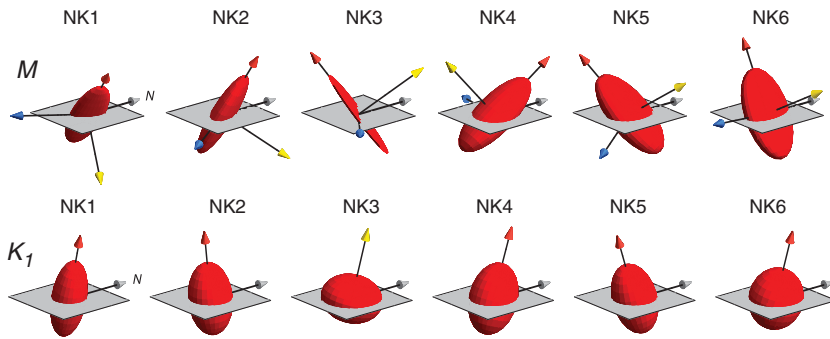
coordinates (γ, δ) as lune longitude γ and lune latitude δ . For the isotropic tensor, $(\gamma, \delta) = (0^\circ, 90^\circ)$, and for the deviatoric tensor $\delta = 0^\circ$. This is a mathematical decomposition that has no physical basis, as far as we know (Tape and Tape, 2013, section 8.3). But it is a useful starting point for examining decompositions on the lune, including more physical ones, discussed next.

The second representation (Fig. 7c) is the classical model of Aki and Richards (1980), elucidated by Dufumier and Rivera (1997) and Tape and Tape (2013). In this model, a full moment tensor can be interpreted as an oblique opening crack. The angle α between the two possible fault planes is also the angle between the slip vector and normal vector. The second parameter ν is the Poisson parameter of the source region and, in principle, is something that could be estimated from a set of moment tensors (Vavryčuk, 2011; Tape and Tape, 2013). To elucidate the classical model, we can remove an isotropic component given by $\sqrt{3}\lambda_2 I$. This amounts to starting at a point on the lune and then descending along a line of constant longitude until reaching the $\lambda_2 = 0$ arc, as illustrated in [Figure S6](#). These focal mechanism plots, shown in the second row of Figure 7c, reveal the two possible fault planes—in general, not perpendicular—and provide a physical perspective for α , the angle between fault planes. What is not depicted is the Poisson parameter for each moment tensor; these are listed in Figure 7a and Table 3. Five of six events are within the allowable range of Poisson values; NK3, with $\gamma > 0$ and $\delta > 0$, lies within an impermissible region ($\nu > 0.5$), at least according to the classical model.

Each of the six events has one of its possible fault planes oriented near horizontal. This is emphasized by plotting a blue dot—representing the fault normal—on each moment tensor in the bottom of Figure 7c. If the near-horizontal plane is assumed to be the fault plane for each event, then each moment tensor can be interpreted as a single process of opening at an oblique angle across the fault.

The third representation (Fig. 7d,e) is a decomposition of the moment tensor into two processes: a double-couple moment tensor plus a tensional or compressional crack tensor. This CDC model was introduced by Minson *et al.* (2007) and elucidated by Tape and Tape (2013).

The primary assumption is that the crack is aligned with a fault plane of the double couple. For each moment tensor, there are two possible CDC decompositions. The moment tensor M can be expressed as (Tape and Tape, 2013)



▲ **Figure 8.** Ellipsoids depicting the moment tensors for the six explosion events (NK1, ..., NK6). The ellipsoid can be thought of as enclosing the all-radially-outward vector fields associated with each moment tensor. (Top) Moment tensors M depicted in Figure 7a. The principal axes are colored as T (red), null (blue), and P (yellow). The north arrow is black and labeled N. (bottom) Crack moment tensors K_1 depicted in Figure 7d (top). Only the primary principal axis is plotted. The color version of this figure is available only in the electronic edition.

$$M(\phi, \zeta) = (\cos \zeta)D + (\sin \zeta)K(\phi), \quad (1)$$

in which D is a double-couple moment tensor, K is a crack tensor, ϕ is the azimuth on the lune (counterclockwise from top), and ζ (0° – 90°) provides a measure of the crack tensor (versus double couple) in M . The CDC is depicted as an arc on the lune between the double couple at center and the crack tensor at the boundary (⊕ Fig. S6). The end points of the arc are given by $\zeta = 0^\circ$ ($M = D$) and $\zeta = 90^\circ$ ($M = K$). The source type of any full moment tensor can be characterized by ϕ and ζ .

Each of the two fault planes of the classical model (Fig. 7c, bottom) serves as the crack plane for one of the two CDC decompositions. The first CDC decomposition in Figure 7d reveals something interesting: the normal to the plane of crack tensor is near vertical, as indicated by the T axis (and P axis, for NK3) near the center of each focal mechanism plot. Blue dots of the crack tensor principal axes in Figure 7d (top) are exactly the same as those in Figure 7c (bottom), where they denote the normal to one of the fault planes. The shape of the radiation pattern for these all-radially-outward moment tensors (K_1) is governed by the three positive eigenvalues. For five events on the left of the lune, the radiation pattern is elongated vertically, like a cucumber; for the one event on the right of the lune (NK3), the radiation pattern is flat like a pancake. The double couple provides a small contribution to the overall moment tensor, as indicated by the relatively small values of M_w for D_1 and the large values of ζ (e.g., $\zeta > 50^\circ$ in Table 3).

Additional information is needed to choose between the two CDC decompositions in Figure 7d,e. Given that all events are assumed to be explosions in the same region, we might choose the CDC decomposition for each event that shares a common physical component such as a fault plane. The CDC results in Figure 7e imply a steeply oriented crack plane K_2 whose strike direction varies considerably among all events. By comparison, the results for K_1 in Figures 7d and 8 show a consistent near-horizontal crack plane for all events—a

model that has been previously advocated for representing explosion events (Patton and Taylor, 2011; Liu *et al.*, 2018). Still, we are reluctant to interpret the double-couple components D_1 , given their differences among all six events and given their uncertainties. Additional data such as structural geological mapping (Massé, 1981), microseismicity (Julian *et al.*, 2010), or satellite-derived ground deformation (Wang *et al.*, 2018) may offer the best hope for constraining the nonisotropic parts of events.

Time Shifts and 1D Structural Models

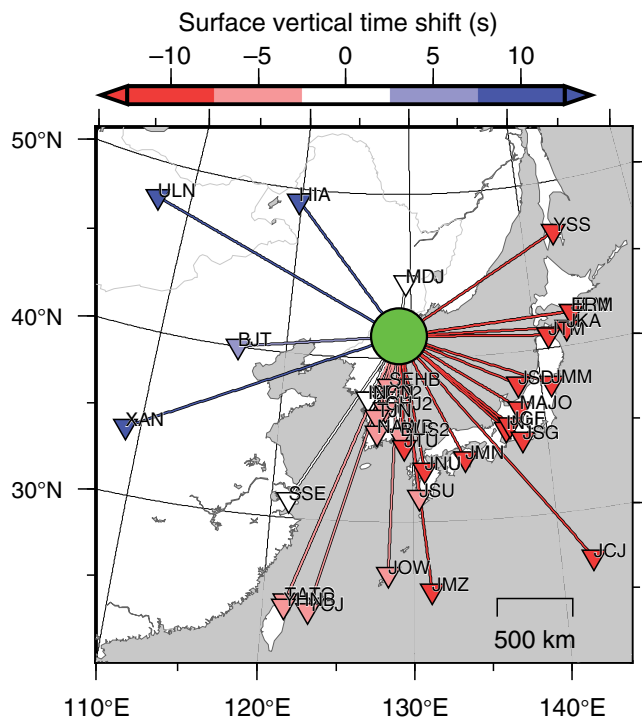
Because of the approximate nature of 1D structural models for modeling seismic waveforms, it is important to carefully analyze and address time shifts between data and synthetics (Silwal and Tape, 2016; Alvizuri *et al.*, 2018). Our analysis of Nevada National Security Site explosions

recorded at distances of > 1000 km showed that time shifts could range by 20 s among different stations: for example, 10 s toward the northwest (Corvallis, Oregon) and -10 s to the southeast (Tucson, Arizona).

For the North Korea events, we use a 1D model (MDJ2) that is calibrated for continental paths, so we might expect large time shifts for the oceanic paths to the east. Figure 9 shows the time shifts for Rayleigh waves for the NK6 best-fitting moment tensor (⊕ Fig. S5). The Rayleigh waves show a striking pattern: to the east, the red (negative) time shifts indicate that the 1D model needs to be sped up significantly; to the west, the blue (positive) time shifts indicate that the 1D model needs to be slowed down. This pattern arises because the Korean peninsula is at the margin of continental crust to the west and oceanic crust to the east in the Sea of Japan (Bassin *et al.*, 2000). For stations within 1000 km, the range in time shifts is about 40 s: from about 5 s to the west (BJT) and -35 s to the east (MAJO, INC). The pattern of time shifts for Love waves is complicated and warrants further study. See ⊕ Figures S7 and S8 for time shift maps of all nine events.

Figure 10 highlights a consequence of time shifts. If there are no restrictions on time shifts, then the synthetic seismograms will be shifted to maximize the cross correlation with the observed seismograms. In some cases, the maximum cross correlation may occur for a synthetic seismogram that is cycle-skipped in its alignment with the data. An extreme case is that if $M(\gamma, \delta)$ is the best-fitting moment tensor with source type (γ, δ) , then $-M$ will provide an identical set of waveforms but with the sign flipped. This moment tensor will appear at $(-\gamma, -\delta)$ on the lune and can provide a local minimum if time shifts are unconstrained. This can be seen by comparing Figure 10a,b. The variance reduction (VR) function $VR(\Lambda)$ on the lune in Figure 10a reveals 180° rotational symmetry about the double-couple point at the center of the lune. For example, the dark blue region of lowest misfit, at the upper left, appears again at the lower right. By carefully limiting the allowable time shifts, we obtain Figure 10b, which no longer exhibits

Event 20170903033001760
Model MDJ2, inv depth 1 km
Freq 0.02 - 0.06 Hz (16.67 - 50.00 s)



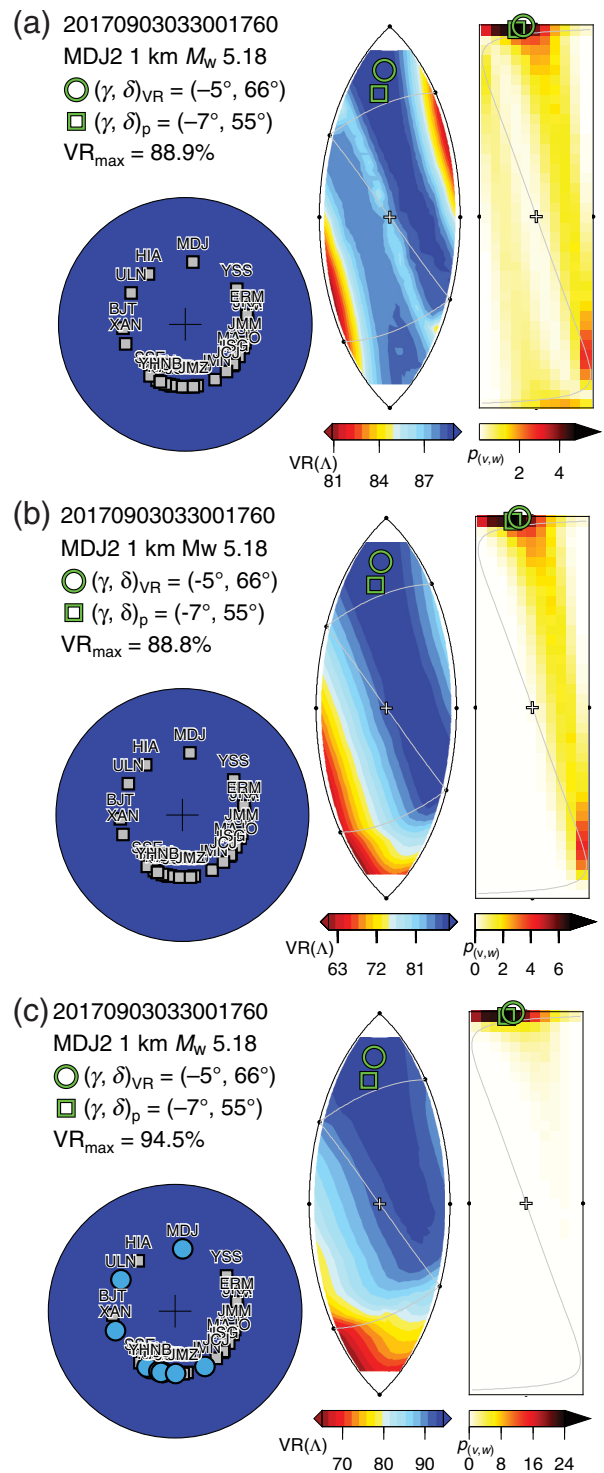
▲ **Figure 9.** Rayleigh-wave time shift maps for the 3 September 2017 explosion event (NK6). The time shifts correspond to the waveform fits in ⑤ Figure S2 (periods 17–50 s). The 1D model we use is calibrated for the path to MDJ, to the north, for which the time shift is 0.7 s. The color version of this figure is available only in the electronic edition.

the rotational symmetry. Identifying the allowable range for time shifts involves examining the unconstrained time shifts for nearby stations for multiple events and then also examining time shifts over multiple bandpass period ranges.

SUMMARY

Using regionally recorded three-component seismic waveforms, we estimated full moment tensors for seven events from North Korea in addition to two comparison events to the south (Fig. 1). We performed a detailed analysis of the largest and most recent event (NK6: 3 September 2017) from the North Korea nuclear test site. Our summary points are as follows:

1. Moment tensor source types for all nine events are separated on the eigenvalue lune (Fig. 6): six explosion events toward the top, two earthquakes in the center, and the one collapse event toward the bottom. This source type discrimination capability for moment tensors was originally demonstrated for the Nevada National Security Site by Ford *et al.* (2009a) and revisited by Alvizuri *et al.* (2018).
2. A four-station analysis for all six explosions provides a reliable relative estimate of moment magnitudes derived



▲ **Figure 10.** The influence of first motion polarities and allowable time shifts on the uncertainty analysis for the 3 September 2017 explosion event (NK6). The maximum VR values are 88.9, 88.8, and 94.5. (a) Without polarities and without custom allowable time shifts. The time shifts between data and synthetics are allowed within ± 50 s for this event. (b) Without polarities, with custom allowable time shifts. (c) With polarities, with custom allowable time shifts. The color version of this figure is available only in the electronic edition.

using regional surface waves filtered between 20–50 s (Table 2).

3. Moment tensors for the six explosion events from the North Korea test site reveal striking similarities (Fig. 3). The best-fitting moment tensors are solid-colored focal mechanism plots, representing first motions that are all radially outward with respect to the source region. We apply the CDC decomposition (Minson *et al.*, 2007; Tape and Tape, 2013), and for each event, we find that the constituent crack tensor has its normal vector pointing upward (Fig. 7d). This is obtained directly from our best-fitting moment tensor and without any assumption of the crack orientation.
4. Our uncertainty characterization follows Alvizuri *et al.* (2018). We convey the uncertainty in moment tensor source type by showing the best moment tensor, over all orientations, for each source type Λ on the lune (Fig. 4). The vw rectangle hosts the probability density $p(v, w)$ for source type. In general, the source type of the maximum of $VR(\gamma, \delta)$, on the lune, is not the same as the maximum of $p(v, w)$, on the rectangle, which takes into account the waveform fits over all orientations. For example, whereas the nonexplosion event in Figure 5b reveals agreement between the two (both near the double couple), the event in Figure 5c shows a $p(v, w)$ maximum having a large negative isotropic component, and the $VR(\gamma, \delta)$ on the lune is almost exactly a double couple. These differences highlight the importance of distinguishing between the best-fitting moment tensor and the most likely source type.
5. The uncertainties in moment tensor source type (on the lune or on the vw rectangle) can be narrowed using first-motion polarities, more stations, and narrower allowable time shifts between data and synthetics (Fig. 10). Our preferred solution for the largest event (NK6) is shown in Figure 4.

DATA AND RESOURCES

The facilities of Incorporated Research Institutions for Seismology (IRIS) Data Services, specifically the IRIS Data Management Center, were used for access to waveforms, related metadata, and derived products used in this study. IRIS Data Services are funded through the Seismological Facilities for the Advancement of Geoscience and EarthScope (SAGE) Proposal of the National Science Foundation under Cooperative Agreement EAR-1261681. 

ACKNOWLEDGMENTS

C. A. was partially supported by the Swiss National Science Foundation (Grant PP00P2_157627: OROG3NY). C. T. was supported by the U.S. Air Force Research Laboratory (Contract FA9453-17-C-0025). The authors thank Editor Bill Walter and two anonymous reviewers for providing thorough and constructive feedback.

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Published Online 19 September 2018